

ADCSLab

REGULATORY SANDBOX

WORKFLOW

Assessments

Define Stress Case

Assessment Plan

Run Assessment

Results

Settings

ADCSL v2.1.0

Sandbox — not for production

Agent-Proposed Assessment Plan

ASS-2024-032 — Equity Flash Crash

1 Define Stress Case

2 Assessment Plan

3 Run Assessment

4 Results

ADCSL Supervisory Agent v2.1

PROPOSAL

The following plan was generated from your stress case definition. It is a proposal only — regulatory review and approval are required before execution. The agent supports the assessment but does not make regulatory determinations.

1 Baseline Calibration

~8 min

Establish the pre-shock operating state for each of the five selected decision systems. Capture current positions, risk parameters, active model versions, configuration flags and any in-flight orders at T-30m.

EVIDENCE REQUIRED

- System configuration snapshots (T-30m)
- Pre-shock position files from each system

KEY ASSUMPTION

All systems are operating in standard production mode at baseline.

2 Shock Injection

~3 min

Administer the identical synthetic equity flash crash stimulus to all five systems simultaneously using a controlled, isolated market data feed. No inter-system information leakage during injection.

EVIDENCE REQUIRED

- Synthetic tick data feed — May 2010 derivative
- Order book reconstruction — NYSE/NASDAQ

KEY ASSUMPTION

Systems cannot observe each other's responses during the injection phase.

3 Response Collection

~60 min (simulated)

Capture decision outputs at five observation intervals: T+0, T+5m, T+15m, T+30m and T+60m. Record order submissions, cancellations, risk limit breaches and liquidity withdrawal events.

EVIDENCE REQUIRED

- Order flow logs
- Risk limit breach event logs
- Liquidity provision/withdrawal records

KEY ASSUMPTION

Simulated time compression — 60-minute window modelled at 10:1 ratio.

4 Directional Convergence Analysis

~12 min

Assess whether systems made aligned directional decisions at each observation interval. Compute pairwise directional correlation, timing spread and aggregate convergence score across all five systems.

EVIDENCE REQUIRED

- Response collection outputs from Step 3

5 Collective Market Impact Estimation

~18 min

Model the aggregate price impact of simultaneous system responses using a synthetic market microstructure model. Compare to counterfactual individual-system impact to quantify the amplification effect of convergence.

EVIDENCE REQUIRED

- Market microstructure model v4.1
- Aggregate liquidity depth estimates

KEY ASSUMPTION

Model calibrated to 2024 equity market conditions, not 2010 conditions.

6 Feedback Loop Assessment

~15 min

Identify second-order amplification dynamics: cases where system A responses alter market conditions that trigger system B, C or D responses, producing a reinforcing cycle.

EVIDENCE REQUIRED

- Collective impact model outputs from Step 5

7 Verification and Cross-Check

~10 min

Cross-reference findings against documented market behaviour from the May 6, 2010 flash crash and ESMA AI convergence research. Flag material divergences from historical patterns as uncertainty factors.

EVIDENCE REQUIRED

- SEC/CFTC Joint Report — September 2010
- ESMA AI Convergence Study 2023

KEY ASSUMPTION

Historical comparison is indicative, not definitive.

VERIFICATION CHECKS

☒

All 5 systems receive identical shock data

☒

No inter-system communication during assessment

☒

Microstructure model version locked at v4.1

☒

Baseline state snapshots validated pre-run

☒

Cross-check against SEC/CFTC 2010 report

☒

Uncertainty bounds computed at each step

PLAN SUMMARY

Total Steps

7

Estimated Duration

~126 min

Systems Assessed

5

Observation Points

5 intervals

Evidence Sources

9 items

Verification Checks

6 checks

OPTIONAL COMPARISON

Add an alternative scenario to run alongside — for example, the same shock with a circuit-breaker safeguard active.

Created with Figma Make

Remix

...

X

+ Add Comparison Scenario

1

 Approving this plan authorises the assessment to run. The agent will execute each step in sequence and surface results for your review.

Edit Stress Case

Approve and Run Assessment →